

D690

Elementary Disciplinary Literacy

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A fourth-grade teacher who recently moved from another district wants students to read a science article “like scientists.” The article is printed on glossy paper and has four photos. Which instruction best reflects disciplinary literacy?

- A. Have students summarize the article in three sentences.
- B. Have students identify the claim, examine the data offered, and note whether the conclusions go beyond the evidence.
- C. Have students discuss which photo is most beautiful.
- D. Use the same questioning routine for every subject, since reading is reading.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

On a worksheet printed on bright yellow paper, a third grader is asked, “What is the product of 6 and 4?” She writes “a toy.” The class had just returned from a field trip to a toy factory. What is the best instructional response?

- A. Mark it wrong and move on; she should know the word.
- B. Treat it as a careless error from the field trip excitement.
- C. Explicitly teach that in math “product” means the result of multiplying, and contrast it with the everyday meaning.
- D. Tell her product means the answer to any math problem.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A teacher wants a list of math words that also have a common everyday meaning. Which list fits best?

- A. quotient, numerator, parallelogram, denominator
- B. Every word in math has two meanings.
- C. volume, table, odd, mean
- D. happy, run, blue, jump

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Fifth graders tested which of three cups kept water warm longest. The foam cup lost the least heat. One student, whose older brother is a chemist, writes: "Foam is the best insulator. Our foam cup dropped only 5 °C in 20 minutes while the metal cup dropped 18 °C." What is missing from the explanation?

- A. Reasoning that connects the data to a science idea, such as foam's trapped air slowing heat transfer.
- B. Evidence.
- C. A quote from an expert, such as a chemist.
- D. A claim.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A student's science claim is "Plants need light to grow." Which sentence is evidence rather than reasoning?

- A. "Plants need light to grow."
- B. "I think plants are cool."
- C. "Plants use light energy to make food through photosynthesis."
- D. "The plant in the window grew 8 cm, but the plant in the closet grew 1 cm."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Fifth graders studying their state's gold rush read a newspaper story, printed in large bold type, claiming a miner found a gold nugget "the size of a horse." The class also has a mining company's records and two miners' letters from the same year. What should students do?

- A. Compare the claim with the company records and letters to see whether they support it.
- B. Summarize the story in their own words.
- C. Accept it, since newspapers report facts.
- D. Accept it, because the bold type shows it was important.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A second grader reading a nonfiction book about frogs, which has a shiny cover and 48 pages, needs to find what “tadpole” means. Which text feature should the teacher point her to?

- A. The glossary.
- B. The table of contents.
- C. Read all 48 pages until she finds it.
- D. The index.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A bar graph in a fourth-grade science magazine shows rainfall for two towns. Town A’s bar looks three times as tall as Town B’s. The vertical axis starts at 40 inches, Town A’s bar reaches 46, and Town B’s reaches 42. The magazine is popular with students. How much more rain did Town A get than Town B?

- A. 6 inches more.
- B. 4 inches more (46 vs. 42).
- C. Three times as much.
- D. 88 inches more.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A student claims a graph proves one plant grew much more than another. What should the student check first?

- A. Whether the graph has a caption with exciting words.
- B. The axis labels, units, and scale — including whether the axis starts at zero.
- C. The colors of the bars.
- D. Nothing; a graph always shows the truth.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Fifth graders researching sharks find a site with a polished design, a .org address, and hundreds of shark photos. The site's home page loads quickly on the school's new laptops. How should the teacher guide them to judge whether it is trustworthy?

- A. Trust it, because it loads fast.
- B. Trust it, because .org sites are always reliable.
- C. Trust it, because the design looks professional and has many photos.
- D. Open new tabs and search for who runs the site and what trusted sources say about that group.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A second-grade class with a new math curriculum solves: "Maya had some stickers. She gave away 5 and has 12 left. How many did she start with?" The problem is printed next to a sticker illustration. A student subtracts because of the word "left." What is the best teaching response?

- A. Accept $12 - 5 = 7$ as the answer.
- B. Praise the student, since "left" always means subtract.
- C. Have the student count the stickers in the illustration.
- D. Have the student model the story with a bar or drawing, showing the unknown start, which leads to $12 + 5 = 17$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Which word problem is most likely to trick a student who relies on the keyword "in all" to mean add?

- A. "There are 10 birds. 3 fly away. How many are left?"
- B. "Lia read 6 books and then 2 more. How many did she read?"
- C. "There are 24 apples in all, packed equally into 4 bags. How many are in each bag?"
- D. "Sam has 3 red cars and 4 blue cars. How many cars in all?"

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A fourth-grade teacher ~~who recently moved from another district~~ wants students to read a science article “like scientists.” ~~The article is printed on glossy paper and has four photos.~~ Which instruction best reflects disciplinary literacy?

Target word: disciplinary approach **Decides:** Teach how scientists read — evidence, data, and precise claims.

A	Adjacent category	A useful general strategy, but not discipline-specific.
B	Correct	Mirrors how scientists read: evaluating claims against data.
C	Camouflage	Uses the photos; not a scientific reading move.
D	Absolute language	Disciplines read and reason differently.

2. Answer C Concept 2

On a worksheet ~~printed on bright yellow paper~~, a third grader is asked, “What is the product of 6 and 4?” She writes “a toy.” The class had just returned from a field trip to a toy factory. What is the best instructional response?

Target word: respond to a math-vocabulary confusion **Decides:** “Product” has a special math meaning.

A	Dropped element	Drops the instruction on the math meaning.
B	Camouflage	The trip explains why the everyday meaning came to mind, but the gap is the math meaning, which still needs teaching.
C	Correct	Addresses the double meaning directly.
D	Adjacent category	Replaces one misconception with another; sum, difference, and quotient differ.

3. Answer C Concept 2 · transfer

A teacher wants a list of math words that also have a common everyday meaning. Which list fits best?

Target word: choose words with dual meanings **Decides:** Words used differently in everyday talk.

A	Inverted operation	Math-only terms without common everyday meanings.
B	Absolute language	Many math terms are used only in math.
C	Correct	Each has an everyday meaning different from its math meaning.
D	Adjacent category	Everyday words with no special math meaning.

4. Answer A Concept 3

Fifth graders tested which of three cups kept water warm longest. The foam cup lost the least heat. One student, ~~whose older brother is a chemist~~, writes: “Foam is the best insulator. Our foam cup dropped only 5 °C in 20 minutes while the metal cup dropped 18 °C.” What is missing from the explanation?

Target word: missing CER part **Decides:** The claim and data are present; the reasoning is not.

A	Correct	Claim and evidence are there; reasoning links them to a principle.
B	Confusion set	The temperature changes are the evidence.
C	Camouflage	Keys on the brother; authority is not reasoning.
D	Dropped element	“Foam is the best insulator” is the claim.

5. Answer D Concept 3 · transfer

A student's science claim is "Plants need light to grow." Which sentence is evidence rather than reasoning?

Target word: identify evidence — reverse of finding the missing reasoning **Decides:** Evidence is data from the investigation.

A	Confusion set	That is the claim itself.
B	Adjacent category	Opinion, not evidence.
C	Inverted operation	A science principle — that is the reasoning.
D	Correct	Measured data from the test.

6. Answer A Concept 4

Fifth graders studying their state's gold rush read a newspaper story, ~~printed in large bold type~~, claiming a miner found a gold nugget "the size of a horse." The class also has a mining company's records and two miners' letters from the same year. What should students do?

Target word: historian's move **Decides:** Check the claim against other sources.

A	Correct	Corroboration tests a claim against other evidence.
B	Adjacent category	A general strategy that skips evaluating the claim.
C	Absolute language	Newspapers can exaggerate, especially to sell papers.
D	Camouflage	Typeface says nothing about accuracy.

7. Answer A Concept 5

A second grader reading a nonfiction book about frogs, ~~which has a shiny cover and 48 pages~~, needs to find what "tadpole" means. Which text feature should the teacher point her to?

Target word: which feature **Decides:** A definition is found in the glossary.

A	Correct	Defines key terms in the book.
B	Adjacent category	Lists chapters, not word meanings.
C	Wrong stem number	Uses the page count; text features exist to avoid this.
D	Confusion set	Lists page numbers where a topic appears, not definitions.

8. Answer B Concept 6

A bar graph in a fourth-grade science magazine shows rainfall for two towns. Town A's bar looks three times as tall as Town B's. The vertical axis starts at 40 inches, Town A's bar reaches 46, and Town B's reaches 42. ~~The magazine is popular with students.~~ How much more rain did Town A get than Town B?

Target word: read the graph correctly **Decides:** The axis starts at 40, so compare actual values.

A	Wrong stem number	Measures Town A's bar from 40 (46 – 40).
B	Correct	$46 - 42 = 4$; the actual difference is about 10%, not triple.
C	Dropped element	Drops the axis starting at 40; bar heights mislead.
D	Inverted operation	Adds the values instead of subtracting.

9. Answer B Concept 6 • transfer

A student claims a graph proves one plant grew much more than another. What should the student check first?

Target word: first check on a graph **Decides:** Title, axes, units, and scale.

A	Confusion set	Captions can mislead; the data show the size.
B	Correct	Scale and units determine what bar sizes really mean.
C	Adjacent category	Color does not change the values.
D	Absolute language	Graphs can be drawn to exaggerate.

10. Answer D Concept 7

Fifth graders researching sharks find a site with a polished design, a .org address, and hundreds of shark photos. ~~The site's home page loads quickly on the school's new laptops.~~ How should the teacher guide them to judge whether it is trustworthy?

Target word: evaluating a website **Decides:** Leave the site and check what others say about who runs it.

A	Camouflage	Speed says nothing about accuracy.
B	Absolute language	Anyone can register a .org address.
C	Dropped element	Drops any check of who is behind it.
D	Correct	Lateral reading checks reputation from outside the site.

11. Answer D Concept 8

A second-grade class ~~with a new math curriculum~~ solves: "Maya had some stickers. She gave away 5 and has 12 left. How many did she start with?" ~~The problem is printed next to a sticker illustration.~~ A student subtracts because of the word "left." What is the best teaching response?

Target word: respond to keyword thinking **Decides:** The start is unknown; modeling shows it is $12 + 5$.

A	Inverted operation	Uses the wrong operation.
B	Absolute language	Keywords fail on start-unknown problems.
C	Camouflage	The picture does not model the problem.
D	Correct	Modeling the situation reveals the structure the keyword hides.

12. Answer C Concept 8 • transfer

Which word problem is most likely to trick a student who relies on the keyword "in all" to mean add?

Target word: find the problem where the keyword misleads **Decides:** "In all" appears but the operation is not addition.

A	Adjacent category	Uses a different keyword that happens to match the operation.
B	Confusion set	No "in all," and addition is correct.
C	Correct	"In all" is the total already given; the operation is division.
D	Inverted operation	Here addition is correct, so the keyword does not mislead.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Disciplinary vs. content-area literacy	Content-area literacy applies general strategies (summarizing, questioning) to any subject; disciplinary literacy teaches how experts in that field read, write, and reason.	Assuming one generic strategy works the same in science, math, and history.	A lesson asking students to read “like a scientist” or “like a historian.”
2	Everyday words with math meanings	Many math words have a different everyday meaning (product, table, volume, odd, mean, difference); they must be taught explicitly in their math sense.	Assuming students know a word because it is common.	A student answer that uses the everyday meaning of a math term.
3	Claim, evidence, reasoning in science	Science writing states a claim, supports it with data, and uses reasoning to connect the evidence to a scientific principle.	Stopping at claim plus data, or offering opinion as evidence.	A science explanation missing the “why the evidence supports it” link.
4	Corroboration in history	Historians check a claim against several sources and note where they agree and disagree before accepting it.	Trusting a single vivid account.	One source with a dramatic claim and others available.
5	Using informational text features	Headings, bold words, captions, glossaries, and indexes help readers locate and understand information; teach students which feature serves which purpose.	Reading every page front to back to find one fact.	A task to find or define something quickly in a nonfiction text.
6	Reading data displays	Before drawing conclusions, read the title, axes, units, and scale of a graph or table.	Reading bar heights without checking the scale or units.	A graph with a scale that does not start at zero or units that change.
7	Lateral reading for digital sources	Judge a website by leaving it and checking what other trusted sources say about who runs it, rather than trusting how the page looks.	Trusting a site because it looks professional or ends in .org.	Students evaluating an unfamiliar website for a report.
8	Reading word problems for structure	Students understand a word problem by modeling its situation (what is known, what is unknown, how the quantities relate), not by hunting for keywords.	Teaching that “left” always means subtract or “in all” always means add.	A word problem where the keyword points to the wrong operation.

Course scope (public description): Elementary Disciplinary Literacy is an innovative course designed for candidates seeking to deepen their understanding and application of literacy skills within specific academic disciplines. This course focuses on the unique literacy demands of content disciplines, exploring how reading, writing, speaking, and listening are used differently in each domain. Students investigate the specialized language structures and text features inherent to each discipline and learn strategies to help learners navigate and master these complexities. The curriculum emphasizes the development of skills through the science of reading that

enable candidates to guide learners in critically engaging with and producing disciplinary texts. Candidates also explore ways to integrate technology and digital literacy into their teaching practices, enhancing students' ability to access, interpret, and share discipline-specific information. Through a blend of research-based evidence and authentic learning activities using structured literacy practices, this course aims to give educators the skills needed to effectively teach disciplinary literacy, thereby improving student achievement and content understanding across all areas of study.